

Carlos De Manuel Vicente

Data & Analytics Engineer

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Data & Analytics Engineer at a global automotive manufacturer. I build the models, pipelines and apps that 20+ plants in North America, APAC and EMEA use every day, from the reports top management reads to the tools people use on the shop floor. Stack: Snowflake, dbt, Python, SQL, Prefect, AWS, Power BI and Streamlit.

EXPERIENCE

Data & Analytics Engineer Jan 2025 – Present
RESRG Automotive · global automotive manufacturer Liria, Spain (hybrid)

- Built Power BI dashboards for production, quality, supply chain, purchasing, finance, HR and EHS across **20+ plants** in NA, APAC and EMEA, from part quality and incidents to material-consumption variances and energy prices, used from top management down to plant teams.
- Developed Streamlit apps for the shop floor and management, including inventory tools and a KPI app where plants manage **130+ KPIs** and their targets, with dbt testing the data written back to Snowflake.
- Automated indicators with Python ETL from REST APIs, email and other sources into Snowflake, orchestrated with Prefect, including near-real-time inventory data. Made stock loading **~90% faster** at **~30% lower cost**, mostly by cutting idle time.
- Deployed the apps on AWS (EC2, S3, EFS) with Docker, CI/CD through Makefiles, and Prometheus for metrics and observability.
- Designed Snowflake data models for apps and reports, and tuned queries and views for performance.
- Forecast plant electricity consumption from production variables for the purchasing team, joining utility invoices with production and financial data so usage normalises against sales.

Data Scientist (internship) Nov 2023 – Jul 2024
Biomedical Data Science Lab, ITACA, UPV Valencia, Spain

- Built an end-to-end Python framework and preprocessing pipelines to evaluate healthcare ML models for fairness, robustness and explainability (XAI).

PROJECTS

Job Market Intelligence · [case study](#) · github.com/carlosdmv7/job-market-intelligence
Reads EU data-job ads every day and ranks them against a CV: six free APIs, LLM enrichment that went from **23 to 180** ads a day on the same free quota, dbt on MotherDuck, Prefect Cloud and Streamlit.

Spanish Housing Radar · [case study](#) · github.com/carlosdmv7/spanish-housing-radar
Finds under-priced flats in València by scoring every listing against its neighbourhood: weekly scraping, dbt with **162 data tests**, Prefect Cloud, Docker and Streamlit.

SKILLS

Warehouse & modelling	Snowflake, dbt, DuckDB, Kimball dimensional modelling
Languages & integration	Python, SQL, Bash, Jinja, DAX, REST APIs
Platform & orchestration	AWS (EC2, S3, EFS), Prefect, Airflow, Docker, Make, GitHub Actions, Git, Azure DevOps
Quality & observability	pytest, Pydantic, ruff, mypy, Prometheus
BI & data apps	Power BI, Streamlit, Microsoft Fabric
AI & ML	Gemini and OpenAI APIs (incl. Azure OpenAI), LLM evals, scikit-learn, SHAP

EDUCATION

MSc, Data Analysis, Process Improvement & Decision Support Engineering · UPV, Valencia
BSc, Biomedical Engineering (ICT specialisation) · UPV, Valencia
Thesis graded 10/10 · distinction for academic performance (top 25%)

LANGUAGES

Spanish (native) · English (C1, Cambridge, 194)